UNIVERSITÄT
LEIPZIG

Annotation and n-gram Models

Natural Language Processing

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ANNOUNCEMENT

Room change for next week (30.04.2026): Lecture in Hörsaal 8!

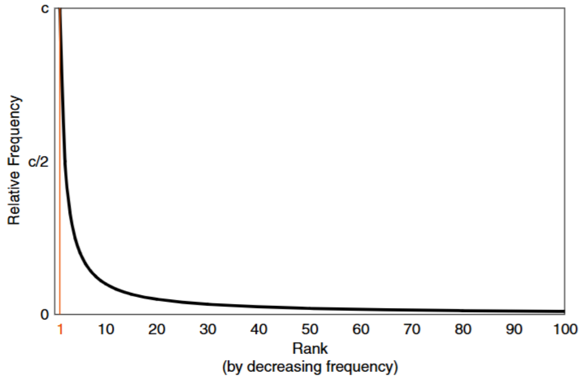
RECAP

TERM FREQUENCY: ZIPF'S LAW

– The relative frequency P (w) of a word w in a sufficiently large text (collection) inversely correlates with its frequency rank r(w) in a power law:

$$P(w) = \frac{c}{r(w)^a} \qquad \Leftrightarrow \qquad P(w) \cdot r(w)^a = c,$$

– where c is a constant and the exponent a is language-dependent; often a ≈ 1.



REGULAR EXPRESSION SYNTAX

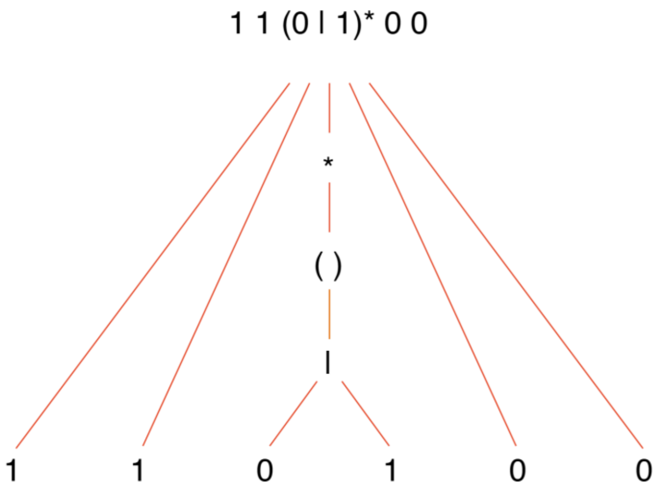
A regular expression R can be composed recursively as follows, where F and G denote regular expressions:

	R	Language $L(R)$	Semantics
1.	a	$\{a\}$	The letter a
2.	FG	$\{fg \mid f \in L(F),\ g \in L(G)\}$	Concatenation
3.	$F \mid G$	$\{f \mid f \in L(F)\} \ \cup \ \{g \mid g \in L(G)\}$	Alternation
4.	(F)	$(L(F))$	Grouping
5.	F^+	$\{f_1 f_2 \ldots f_n \mid f_i \in L(F),\ n \geq 1,\ i = 1, \ldots, n\}$	Non-empty sequence of words from $L(F)$
6.	F^*	$\{\varepsilon\} \ \cup \ L(F^+)$	Arbitrary sequence of words from $L(F)$
7.	F^n	$\{f_1 f_2 \ldots f_n \mid f_i \in L(F),\ i = 1, \ldots, n\}$	Sequence of n words from $L(F)$
8.	ε	$\{\varepsilon\}$	The empty word

EXAMPLES

Every word in the language of this regular expression consists of two ones, followed by any number of zeros or ones, followed by two zeros.

- Concatenation (5 times)
- Sequence
- Bracketing
- Alternative
- Symbol



MORE EXAMPLES

R	Name of language $L(R)$	Words from $L(R)$
(a | b) (c | d | ε)	*Abc*	ac, bc, ad, bd, a, b
Dear(ε | est) (Sir | Madam)	*Salutation*	Dear Sir
0 | 1 | . . . | 9	*Digit*	7
a | b | . . . | z	*sLetter*	x
A | B | . . . | Z	*cLetter*	B
sLetter | *cLetter*	*Letter*	m, N
Letter (*Letter* | *Digit*)*	*Label*	Maximum, min7, a
$Digit^{+}.Digit^{2}$	*MoneyAmount*	23.95, 0.50
$(cLetter \mid cLetter^{2} \mid cLetter^{3})$– $(cLetter \mid cLetter^{2})$– $(Digit \mid Digit^{2} \mid Digit^{3} \mid Digit^{4})$	*LicensePlatesDE*	PB–AS–0815
$1^{3}\ (1 \mid 0)^{*}\ 0^{3}$	*Dual*	1111000, 1111101010000

PERL COMPATIBLE REGULAR EXPRESSIONS

A regular expression R can be composed recursively as follows. Let F and G denote regular expressions and L(F), L(G) their languages over the alphabet Σ.

	R [pcre.org]	Semantics
1.	a	The letter a
	.	Any symbol from Σ (except line break)
	$[\Sigma']$	Character class: Any symbol from $\Sigma' \subseteq \Sigma$
	$[\hat{}\Sigma']$	No symbol from character class $\Sigma' \subseteq \Sigma$
	$\backslash\Sigma'$	Escaped letter from a controlled $\Sigma' \subset \Sigma$, denoting a character class
2.	FG	Concatenation
3.	$F \mid G$	Alternation
4.	(F)	Grouping or capturing
	$F?$	F is optional (same semantics as $F \mid \varepsilon$)
5.	$F+$	Non-empty sequence of words from $L(F)$
6.	$F*$	Arbitrary sequence of words from $L(F)$
7.	$F\{n\}$	Sequence of n words from $L(F)$
	$F\{m, n\}$	Sequence with at least m and at most n words from $L(F)$
	ˆ or $	Assertion: Beginning or end of the text string (nothing may precede or follow it)

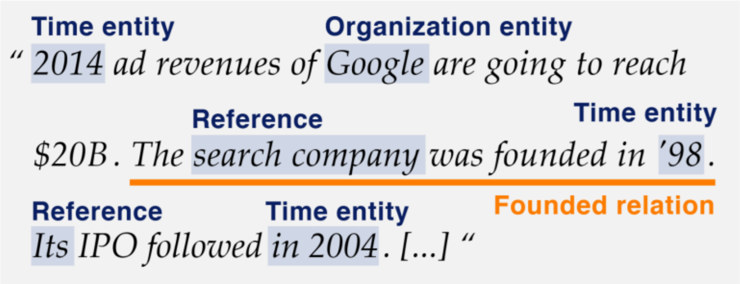
NOT IN THE EXAM

- no atomic groups
- no lookarounds
- \n syntax only for group referencing

DATA ANNOTATION

TYPES OF ANNOTATIONS

Annotation is the process of adding interpretative, linguistic information to a corpus.



Topic: ”Google revenues“ Genre: ”News article“

Types of annotations:

- Classes or labels that interpret entire documents or text spans within
- Derived texts of documents or text spans within (e.g. summaries, image descriptions)
- Relations between annotations

ACQUIRING ANNOTATIONS

Annotation is the process of adding interpretative, linguistic information to a corpus.

Acquiring annotations:

- Manual. Annotations added by humans. Often called ground truth or gold standard.
- Automatic. Annotations from models or algorithm. Sometimes called silver standard.
- External. Annotations obtained with the data, either user-generated or automatic.

ANNOTATION QUESTIONS

Annotation is the process of adding interpretative, linguistic information to a corpus.

Questions:

- How can we encode annotations in a corpus?
- How can we collect the annotations?
- How can we ensure annotation quality?

EXAMPLE: SANTA BARBARA CORPUS OF SPOKEN AMERICAN ENGLISH (SBCSAE)

Start	End	Speaker	Intonation unit [source] [transcript] [audio]
293.47	296.77		... <PAR<P what was I gonna say.
296.77	297.67		.. I forgot what I was think- --
297.67	299.69	LENORE:	You sai[d you never] made the horseshoes,
298.42	298.81	LYNNE:	[gonna say] P>PAR>.
299.69	299.90	LENORE:	but,
299.90	301.07	LYNNE:	... (H) Well,
301.07	302.07		% .. %w- u=m,
302.07	304.37		%= when we put em on a horse's hoof,
304.37	305.22		all we do,
305.22	306.88		(H) they're already made.
306.88	307.93		.. they're round.
307.93	309.43		.. we pick out a size.
309.43	309.85		.. you know we'd,
309.85	311.11		like look at the horse's hoof,
311.11	311.44		and say,
311.44	311.98		okay,
311.98	313.43		(H) this is a double-aught.

EXAMPLE: LANCASTER-OSLO/BERGEN CORPUS (LOB)

Doc.	Line	
A07	94	|^And, of course, 29-year-old Gerry, to whom \0Mme Kilian Hennessy
A07	95	has remained so loyal, will continue to partner him henceforth.
A07	96	|^Problem horse Mossreeba even defied Johnny Gilbert's skill in the
A07	97	Metropolitan Hurdle.
A07	98	**[BEGIN INDENTATION**]
A07	99	|^He struck the front after jumping the last but as Keith Piggott
A07	100	says: ^*"He'll come and beat *1anything, *0 but as soon as he gets his
A07	101	head in front up it goes*- and he doesn't want to know.**"
A07	102	**[END INDENTATION**]

Symbol	Semantics
*" **"	Typographic quotes
*-	Dash
*1 *0	Italics
^	New sentence
\0	Abbreviation
|	New paragraph

EXAMPLE: BROWN CORPUS

```
|SN12:30 the_AT fact_NN that_CS Jess's_NP$ horse_NN had_HVD not_* been_BEN
         returned_VBN to_IN its_PP$ stall_NN could_MD indicate_VB that_CS
         Diane's_NP$ information_NN had_HVD been_BEN wrong_JJ ,_, but_CC Curt_NP
         didn't_DOD* interpret_VB it_PPO this_DT way_NN ._.
|SN12:31 a_AT man_NN like_CS Jess_NP would_MD want_VB to_TO have_HV a_AT ready_JJ
         means_NNS of_IN escape_NN in_IN case_NN it_PPS was_BEDZ needed_VBN ._.
```

EXAMPLE: SUSANNE CORPUS

ID	POS	Word	Lemma	Grammar	Syntax explanation
N12:0280.42	AT	The	the	[O[S[Ns:s.	[paragraph [finite clause [noun phrase, singular : logical subject
N12:0290.03	NN1n	fact	fact	.	
N12:0290.06	CST	that	that	[Fn.	[nominal clause
N12:0290.09	NP1f	Jess	Jess	[Ns:S[G[Nns.Nns]	[NP, singular : surface subject of Fn clause [genitive phrase [proper name
N12:0290.12	GG	+<apos>s	–	.G]	
N12:0290.15	NN1c	horse	horse	.Ns:S]	
N12:0290.18	VHD	had	have	[Vdefp.	[verb group, past tense, negative, perfect, passive
N12:0290.21	XX	not	not	.	
N12:0290.24	VBN	been	be	.	
N12:0290.27	VVNv	returned	return	.Vdefp]	
N12:0290.30	IIt	to	to	[P:q.	[prepositional phrase : directional
N12:0290.33	APPGh1	its	its	[Ns.	[noun phrase, singular
N12:0290.39	NN1c	stall	stall	.Ns]P:q]Fn]Ns:s]	
N12:0290.42	VMd	could	can	[Vdc.	[verb group, past tense, modal
N12:0290.48	VV0t	indicate	indicate	.Vdc]	
N12:0300.03	CST	that	that	[Fn:o.	[nominal clause : logical object
N12:0300.06	NP1f	Diane	Diane	[Ns:s[G[Nns.Nns]	[noun phrase, singular [genitive phrase [proper name
N12:0300.09	GG	+<apos>s	–	.G]	
N12:0300.12	NN1u	information	information	.Ns:s]	
N12:0300.15	VHD	had	have	[Vdfb.	[verb group, past tense, perfect, main verb "BE"
N12:0300.18	VBN	been	be	.Vdfb]	
N12:0300.21	JJ	wrong	wrong	[J:e.J:e]Fn:o]	[adjective phrase : predicate complement subject
N12:0300.24	YC	+,	–	.	
N12:0300.27	CCB	but	but	[S+.	[surface subject
N12:0300.30	NP1m	Curt	Curt	[Nns:s.Nns:s]	[proper name : logical subject
N12:0300.33	VDD	did	do	[Vde.	[verb group, negated
N12:0300.39	XX	+n<apos>t	not	.	
N12:0300.42	VV0v	interpret	interpret	.Vde]	
N12:0310.03	PPH1	it	it	[Ni:o.Ni:o]	[pronoun noun phrase : object
N12:0310.06	DD1i	this	this	[Ns:h.	[noun phrase adjunct : manner
N12:0310.09	NNL1n	way	way	.Ns:h]S+]S]	
N12:0310.12	YF	+.	–	.	

AUTOMATIC ANNOTATION

Automatic annotations are cost-effective and enable large corpora.

- Self-supervision: The annotations are part of the original data. (e.g., language modelling)
- Semi-supervision: The annotations for new data are derived from already annotated data.
- Weak or distant supervision: The annotations are derived from relations between the data and external knowledge. (e.g., sentiment from user ratings, entity relations from databases)
- Simulated annotators via LLMs: e.g., LLMs already outperform crowd workers in some text generation tasks)

Automatic annotations are often noisy and must be filtered or cleaned to improve the quality.

MANUAL ANNOTATION

- Experts: Annotations are done by experts trained for the task and in the general area of the annotation (e.g., linguistics, medicine, etc.)
- Laypeople: Training and supervising laypeople on a task.
- Crowdsourcing: Using a platform to recruit click workers with little training or supervision. Easy to recruit many annotators, but needs good task design and evaluation for results of good quality.

The suitability of either approach depends on budget and availability.

MANUAL ANNOTATION: TOOLS

Prodigy

- NLP-focused tool with a deep integration of spaCy, LLMs, and active learning support. Allows custom templates via html
- Expensive licence

Label Studio

- General tool with templates for many tasks, some options for task design
- Free version has limitations, difficult to use in automated workflows (e.g., active learning)

Doccano

- Open source, but quite limited in features

→ Never implement your own annotation tool without a very good reason!

MANUAL ANNOTATION: CROWDSOURCING

Crowdsourcing refers to techniques using collective intelligence:

- Distribute annotation work to many independent annotators.
- Use individual expertise on small parts of the whole. Wikipedia, OpenStreetMap, . . .
- “Wisdom of the crowds”: even if individual assessments vary, the average is often close to the truth.
  - Francis Galton’s Ox: Francis Galton had villagers guess the weight of an Ox after butchering. The median guess was 1208 pounds, and the dressed weight of the ox 1197 pounds.
- Diversify the pool of annotators

MANUAL ANNOTATION: CROWDSOURCING

Crowdsourcing is suitable for data labelling when:

- many examples are required
- the task can be split up and parallelized
- the individual annotations require little training
- the results can be averaged across annotators

Approaches

- Crowdsourcing by voluntary work (e.g., Wikipedia, OpenStreetMap)
- Crowdsourcing by task combination (e.g., reCAPTCHA)
- Crowdsourcing by gamification (e.g., Google Image Labeller)
- Crowdsourcing by paid labour

MANUAL ANNOTATION: CROWDSOURCING PLATFORMS

Amazon Mechanical Turk

- For microtasks (a few seconds up to minutes)
- Supports many (100–10,000), but often low skilled workers
- Allows custom templates via html and javascript

UpWork

- For recruiting experts and specialists
- Usually more expensive

And several more

MANUAL ANNOTATION: CROWDSOURCING ISSUES

- Recruitment: Find annotators with a specific background (e.g., experience, location, language)
- Qualifications: Train annotators and test their abilities to do the task
- Quality Control: Test annotations for correctness. Improve correctness via task design
- Reputation: Good annotators more often take tasks from reputable organisers. Be fair and pay annotators well and in-time.
- Payment: Low pay has adverse effects: poor quality annotations, market degradation, research ethics. →Time the tasks and pay minimum wage

ANNOTATION TASK DESIGN

Requirements

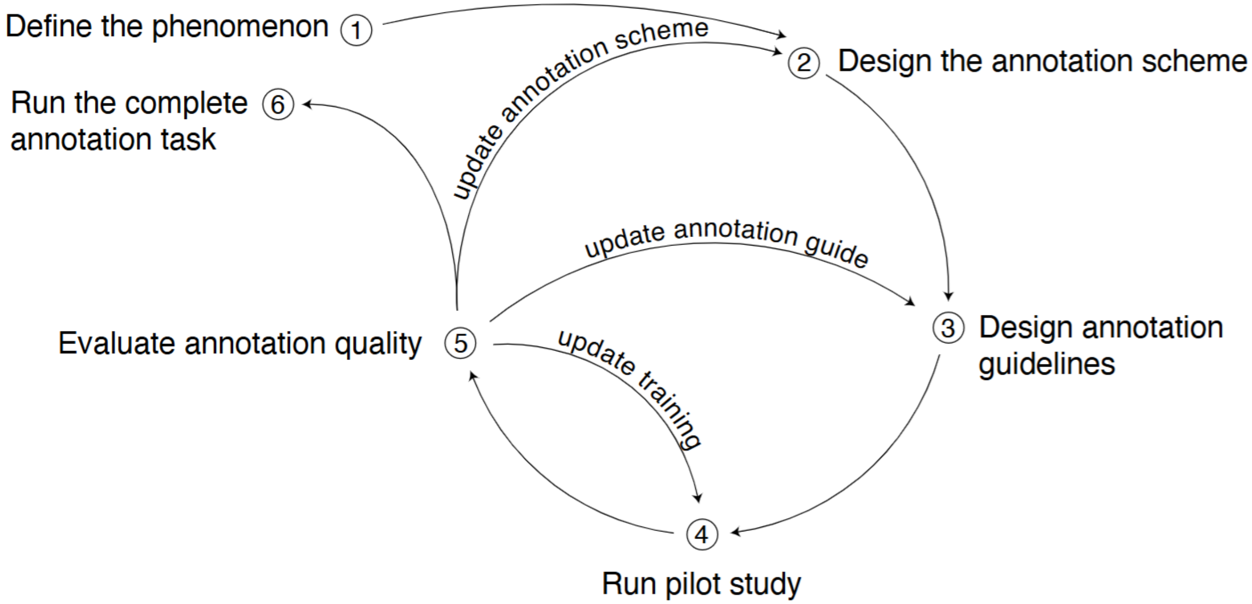
- Correct: The annotations can be trusted, (e.g., experts have high agreement)
- Reproducible: Annotators produce the same annotations when repeating the task

Challenges

- Disagreement: In many cases, there are different beliefs of what is a valid annotation
- Budget, size, and correctness trade-off: Different annotation strategies trade correctness against size. Some noise is acceptable for many projects

ANNOTATION TASK DESIGN

Designing annotation tasks is an iterative process, similarly to software development.



ANNOTATION TASK DESIGN: SCHEMES

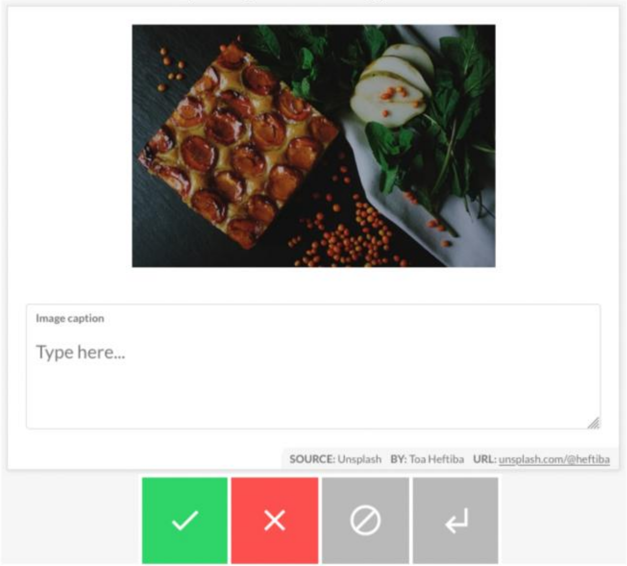
The annotation scheme describes the form (layout) and scope (options) of the annotation task.

Text classification (sentiment)

Span annotation (NER)

Span annotation (entity relations)

Freeform text (image labeling)



ANNOTATION TASK DESIGN: GUIDELINES

Annotation guidelines are the instructions given to the annotators.

Elements of annotation guidelines:

- Definitions of task and the phenomena
- Definitions of annotation options (classes, . . .)
- Typical examples
- Edge cases

Example: Clickbait in microblogs

How Click Baiting Are The Tweets Below?

Clickbait Definitions

“ A tweet is Clickbait if (1) the tweet withholds information required to understand what the content of the article is; and if (2) the tweet exaggerates the article to create misleading expectations for the reader. (cf. Facebook)

“ Clickbait is saying "this town" or "this state" or "this celebrity" instead of saying Los Angeles or Colorado or Justin Timberlake. It's over-promising and under-delivering. It's leaving out the one crucial piece of information the reader may want to know. (cf. HuffPoSpoilers)

“ Clickbait tweets typically aim to exploit the readers curiosity for clicks. They provide just enough information to make readers curious, but not enough to satisfy their curiosity without clicking through to the linked content. (cf. Wikipedia)

Examples

Not Click Baiting	Heavily Click Baiting
David Bowie, the British singer and famous actor, dies aged 69 Link	You'll never believe who tripped and fell on the red carpet... Link
Biggest known example of 'Giant Huntsman Spider' found in Queensland, Australia Link	These heartbreaking wishes of children will change your life Link

Important Notes

- **Don't confuse clickbait with irrelevance!** Just because the tweet is uninteresting or gossip shouldn't imply heavily click baiting.
- **Pay attention to the images!** They might provide information the text misses and reduce how click baiting a tweet is.
- **To prevent abuse**, we manually review and, if apparent, reject assignments. If you are unsure about your performance, do a hand full of HITs and wait for our feedback. We aim to approve within one working day.

ANNOTATION TASK DESIGN: DISAGREEMENT

Annotators make different decisions on the same text in an annotation task.

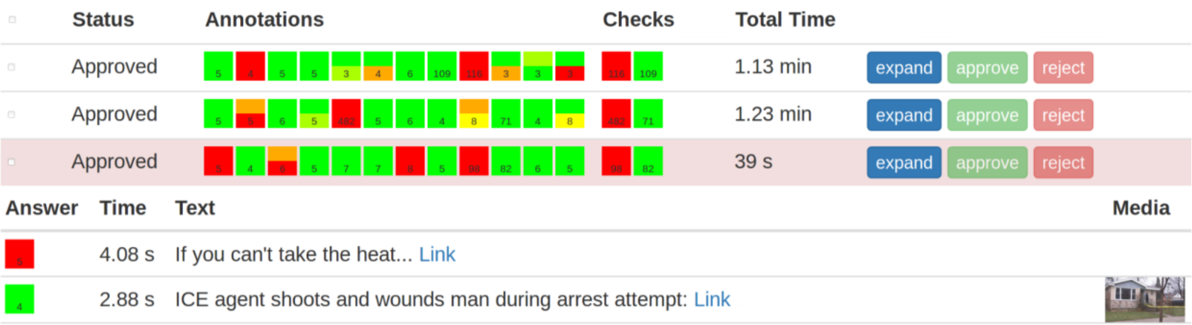
Causes of disagreement:

- Carelessness because of low pay, no consequences, high volume, unclear tasks
- Ambiguity. Misunderstanding content because of metaphors, irony, word plays, . . .
  `“Who knew a side effect of COVID would be gross incompetence.”`
- Missing context: `“Dude this guy is serious? And trump retweeted this?????? Please anonymous take them out”`
- Subjectivity disagreement due to the annotators’ identity, beliefs, and background.
  `#DemocratsAreDestroyingAmerica`

DEALING WITH CARELESS ANNOTATORS

Evaluate and filter

- Check instances: Add some clear and easy examples with known annotations. Remove annotators that get those examples wrong.
- Agreement with other annotators
- Attention checks: Add simple off-topic questions to motivate annotators to pay attention. Raise your hand if you’re still paying attention!
- Dwell time



DEALING WITH CARELESS ANNOTATORS

Evaluate and filter

Make recruitment more restrictive

- Require more experience, more qualifications, ...
- For subjective tasks: restrictive criteria(area, language skills) might reduce diversity and add biases

Options for annotator qualifications on AMT

DEALING WITH DISAGREEMENT

- Vote aggregation (3, 5, . . . votes)
  - Collect multiple annotations for each example and aggregate them.
  - Good for classification, difficult for span annotation or freeform text
  - Means of vote aggregation:

	Data	Use Case
Majority / Mode	nominal*	Select the class with the most votes.
Mean	interval	Select the class closest to the average.
Median	ordinal	Select the class in the middle after ordering.
Minority / Threshold	binary	At least τ positive votes $\rightarrow$ positive example.

DEALING WITH DISAGREEMENT

- Vote aggregation (3, 5, . . . votes)
- Review (2 votes)
  - A layperson annotates, an expert reviews and corrects
  - When annotations are labour intensive (span annotation)
  - When there are many difficult edge cases
- LLM augmentation
  - LLM cast the tie when two annotators disagree
  - LLM decides when an expert needs to review
- Learning with disagreement
- Developing a more prescriptive schema

NON-TECHNICAL ASPECTS

There are ethical and legal considerations when working with human-created data. If in doubt: consult the ethics board before starting an annotation project

- Personal data: Annotations can count as or contain personal data

  → Anonymize and request permission for use

- Legal: Be mindful of data collection and distribution laws and licenses.
- Harmful text: Make annotators aware of potential harm beforehand
- Working Conditions: Provide compensation. Collect and implement feedback

ANNOTATOR AGREEMENT

THE ANNOTATION TASK

Setting of the real world:

- O is a set of objects, e.g. emails
- $S \subset O$ is a sample of objects, e.g. corpus of emails with labels
- C is a set of classes, e.g. spam vs. legitimate
- $\gamma : O \to C$ is a human expert, e.g. a person trained in recognising spam emails

Annotation task: For each $o \in S$, determine its class $\gamma(o) \in C$

- How valid are the annotations $\gamma()$?
- How reliable are the annotations $\gamma()$?

REMARKS

- If a human expert is employed to “compute” γ() for a sample of objects (=annotation), a potential source of errors (label noise) is introduced, for instance, due to lack of expertise (i.e., knowledge or experience) about the target concept, due to honest mistakes, deliberate falsification, unwitting bias, or opinionated interpretation.
- In natural language processing, most annotation tasks are inherently complex, require fine interpretation, or are subjective in nature, increasing the risk that a human expert may randomly or systematically deviate from the target concept.
- Another source of errors for a human expert may be no fault on their part, but due to an ill-defined targetconcept. For instance, the presumed existence of a language phenomenon may be untrue, or its description unsound.

VALIDITY

The quality of being well-founded on fact, or established on sound principles, and thoroughly applicable to the case or circumstances.#

Validity in NLP:

- Establishing validity requires independent sources of evidence of the target concept
- Beyond syntactical concepts, annotating semantic or pragmatic concepts requires interpretation
- As texts are written by someone for someone, it may have meaning to them much more than to an annotator
- Thus, an inherent subjectivity can be associated with annotation
- It is therefore necessary to investigate if annotation can be done at all

RELIABILITY

The degree to which repeated measurements of the same subject under identical conditions yield consistent results.

Types of reliability:

Type	Agreement	Responds negatively to	Strength
Stability	Repetitions (test–retest)	Intra-annotator disagreements	Weakest
Replicability	Replications (test–test)	the above + Inter-annotator disagreements	Intermediate but easily assessed
Accuracy or Surrogacy	Correspondence (test–standard)	all of the above + Deviations from chosen standard	Strongest but standards are not always available

RELIABILITY

Annotation validity → Annotation reliability

- If annotators produce consistently similar results, they may have internalized a similar understanding of the annotation guidelines, and we can expect them to perform consistently under this understanding
- If annotators are not consistent then either some of them are wrong or else the annotation scheme is inappropriate for the data
- Achieving good agreement cannot ensure validity: two observers of the same event may well share the same prejudice while still being objectively wrong

AGREEMENT MEASURES

Annotation setting:

- $S = \{o_1, . . . , o_n\}$ is a set of objects
- $C = \{c_1, . . . , c_k\}$ is a set of classes
- $A = \{\gamma_1, . . . , \gamma_m\}$ a group of human annotators

Analysis task: Measure the agreement of annotators A in classifying the objects S into the classes C.

TERMS FOR AGREEMENT MEASURES

Measures of agreement of human experts on classification task got many different names throughout disciplines and the literature, including different combinations of these terms:

[inter- | intra-]

[analyst | annotator | coder | expert | interpreter | interviewer| judge | observer | rater | scorer | transcriber | unitizer]

[(dis-)agreement | reliability]

OBSERVED AGREEMENT

Let S be a sample of n objects, C a set of k classes, and γ_1 and γ_2 two independent annotators.

The observed agreement is the ratio of objects to which γ_1 and γ_2 assign the same class:

$$a_{\text{obs}} = \frac{|\{o \mid o \in S : \gamma_1(o) = \gamma_2(o)\}|}{n}.$$

Let C_γ denote a random variable assumed to govern the annotations of an annotator γ.

- k = 2 classes: $P(C_\gamma = c_1) = 0.5$ yields $a_{\text{obs}} = 0.50$, and $P(C_\gamma = c_1) = 0.9$ yields $a_{\text{obs}} = 0.82$.
- Bias in favour of fewer classes.
- No correction for class imbalance.
- Non-comparable across annotations with different classes for the same target concept.

CHANCE AGREEMENT

Let S be a sample of n objects, C a set of k classes, and γ_1 and γ_2 two independent annotators.

The observed agreement is the ratio of objects to which γ_1 and γ_2 assign the same class:

$$a_{\text{obs}} = \frac{|\{o \mid o \in S : \gamma_1(o) = \gamma_2(o)\}|}{n}.$$

Expected agreement. Probability that annotators γ_1 and γ_2 assign the same class:

$$a_{\text{exp}} = \sum_{c \in C} P(\boldsymbol{C}_{\gamma_1} = c) \cdot P(\boldsymbol{C}_{\gamma_2} = c)$$

COHEN'S K

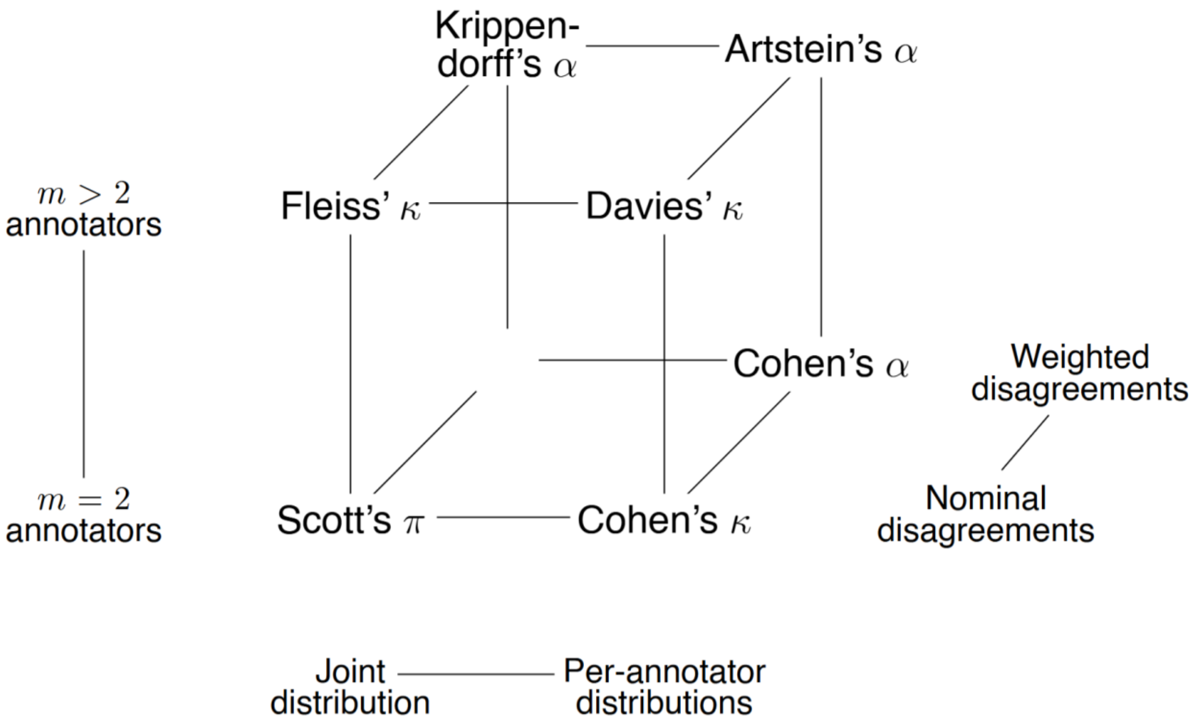
Chance-corrected agreement coefficient:

$$\kappa = \frac{a_{\text{obs}} - a_{\text{exp}}}{1 - a_{\text{exp}}} = \frac{\text{observed agreement above chance}}{\text{potential agreement above chance}}.$$

Range of κ:

- κ = 1 indicates perfect agreement.
- κ = 0 indicates random agreement.
- κ < 0 indicates systematic disagreement.
- The lower bound for κ depends on the specific annotation scenario and approaches −1

OTHER AGREEMENT MEASURES NOT COVERED HERE



QUESTIONS?

N-GRAM LANGUAGE MODELS

INTRODUCTION

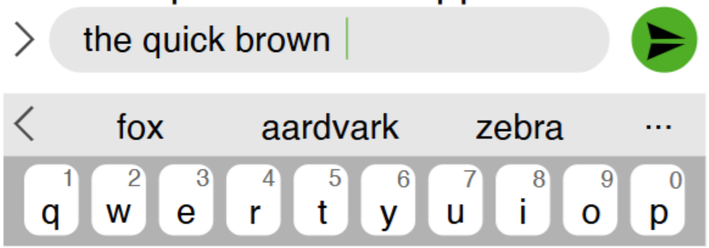
- A language model is a probability distribution over all sentences $(x_1, \ldots, x_n) \in V^*$ with length 1,...,n supported by a fixed-size vocabulary V, with:
  - $\forall (x_1, \ldots, x_n) \in V*: p(x_1, \ldots, x_n) \geq 0$
  - $\sum_{(x_1, \ldots, x_n) \in V^*} p(x_1, \ldots, x_n) = 1$
- A language model can determine the probability of a sentence under V^*
  - Language identification. Is a text more likely under the Spanish or the German language model?
  - Authorship analysis. Estimate one model for each author, predict a disputed text.
  - Information retrieval. Estimate one model for each document, predict the query.

ESTIMATING AND PREDICTING

Estimate the probability of a token $x_n \in V$ following a given sequence $(x_1,\ldots,x_{n-1})$.

1. Estimate $p(x_n)$. $p(x_n \mid x_1,\ldots, x_{n-1})$ conditional probabilities
2. Predict x_n. $x_n = \text{argmax } x \in V\ p(x|x_1,\ldots,x_{n-1})$ greedy decoding

Autocomplete in chat applications



[Guardian, 08 Sep. 2020]


A robot wrote this entire article. Are you scared yet, human?
GPT-3
We asked GPT-3, OpenAI's powerful new language generator, to write an essay for us from scratch. The assignment? To convince us robots come in peace

REMARKS

- Modelling $P(x_n | x_{<n})$ is sometimes called autoregressive or causal language modelling.
- The predicted sentence probabilities depend on the data used to construct the language model.

MAXIMUM LIKELIHOOD LANGUAGE MODELS

A language model can be estimated via maximum likelihood estimation (MLE).

– The probability p(·) of a sentence is its likelihood across all observed sentences. C(·) is the count(·) function.

$$p_{\text{MLE}}(x_1, \ldots, x_n) = \frac{\text{C}(x_1, \ldots, x_n)}{\sum_{\langle x_1, \ldots, x_k \rangle \in V^*} \text{C}(\, x_1, \ldots, x_k \,)}$$

– MLE $(x_n | x_{<n})$ is the likelihood of $p(x_1, \ldots, x_n)$ compared to sequences with the same prefix $x_{<n}$.

$$p_{\text{MLE}}(x_n \mid x_{<n}) = \frac{\text{C}(x_{<n}, x_n)}{\sum_{x_i \in V} \text{C}(x_{<n}, x_i)} = \frac{\text{C}(x_{<n}, x_n)}{\text{C}(x_{<n})}$$

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the quick brown fox jumps over the lazy dog

$x_{<n}, x_n$	$\mathsf{C}(x_{<n})$	$\mathsf{C}(x_{<n}, x_n)$	$p(x_n \mid x_{<n})$
the quick	2	1	1/2 = 0.5
the quick brown	1	1	1/1 = 1
the quick brown fox	1	1	1/1 = 1

Problems?

- We cannot estimate unobserved sequences, independently of their probability
- Most sequences with n>5 are never observed. → Zipf’s Law

AUTO-REGRESSION

Model the probability of a sentence as a series of next-word predictions.

1. Model sentence probability with conditional probabilities: multiplication rule

$$p(x_1,\ldots,x_\mathrm{n}) = p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1,x_2) \cdot \ldots \cdot p(x_\mathrm{n}|x_{<\mathrm{n}}) = \prod_{\mathrm{k}=1}^{\mathrm{n}} p(x_\mathrm{k}|x_{<\mathrm{k}})$$

2. Markov assumption: Limit the condition of each probability to length τ :

$p(x_\mathrm{k}|x_{<\mathrm{k}}) \approx p(x_\mathrm{k}|x_{\mathrm{k}-\tau},\ldots,x_{\mathrm{k}-1})$. This example uses τ = 1

p(x_1, ..., x_n) = p(the, quick, brown, fox, jumps, over, the, lazy, dog)

= p(the | < s >)

· p(quick | the)

· p(brown | quick)

· . . .

· p(dog | lazy)

REMARKS

- Notation: The n in n-gram is not the same n we use as counter for length of the input sequence. The term n-gram can be considered a proper noun.
- The condition for Bayes' theorem is that the predicted events are mutually exclusive. Hence, we must assume that words in a sequence are independent, which is not true.
- The language model parameters can also be estimated through other methods, like Bayes' rule, but MLE is the most common.
- When building language models, each sequence is prepended with a start-of-sequence (<s>) and end-of-sequence token (</s>).
- If a sequence has a token that is not in the vocabulary of the LM, the model cannot estimate the sequences' probability. This can be solved by reducing the vocabulary while fitting the model and replacing all out-of-vocabulary words with a special token <unk>.

BI-GRAM MODEL

Bi-gram Model: a language model under first-order Markov assumption (τ = 1)

- Count all words and word bi-grams in the dataset
- Estimate conditional probabilities via MLE

s_1: <s> I am Sam </s>
s_2: <s> Sam I am </s>
s_3: <s> I do not like green eggs and ham </s>

	<s>	I	am	Sam	do	</s>
<s>	0	0.67	0	0.34	0	0
I	0	0	0.67	0	0.34	0
am	0	0	0	0.5	0	0.5
Sam	0	0.5	0	0	0	0.5
do	0	1	0	0	0	0

$$p(\texttt{I} \mid \texttt{<s>}) = \frac{2}{3} = 0.67$$

$$p(\texttt{Sam} \mid \texttt{am}) = \frac{1}{2} = 0.50$$

$$p(\texttt{do} \mid \texttt{I}) = \frac{1}{3} = 0.34$$

HIGHER ORDER N-GRAM LANGUAGE MODELS

Higher τ can increase text quality

Shakespeare language models:

2-gram Why dost stand forth thy canopy, forsooth; he is this palpable hit the King Henry. Live king. Follow.

3-gram Fly, and will rid me these news of price. Therefore the sadness of parting, as they say, ’tis done.

4-gram King Henry. What! I will go seek the traitor Gloucester. Exeunt some of the watch. A great banquet serv’d in; It cannot be but so.

2-gram	am	I	do	Sam	not
<s>	0	.67	0	.34	0
I	.67	0	.34	0	0
am	0	0	0	.5	0

3-gram	am	I	do	Sam	not
<s> I	.5	0	.5	0	0
<s> Sam	0	1	0	0	0
I am	0	1	0	.5	0

4-gram	am	I	do	Sam	not
<s> I am	0	0	0	1	0
<s> Sam I	1	0	0	0	0
<s> I do	0	0	0	0	1

REMARKS

- n-gram language models are sometimes called maximum entropy models.
- In practice, the log likelihood replaces the probability for numerical stability.

$$\prod_{k=1}^{n} p(x_k \mid x_{<k}) = \text{exp}(\sum_{k=1}^{n} \text{log}(p(x_k \mid 1, \ldots, x_{k-1})))$$

- Determining the probability again (exp) is rarely necessary.
- Entropy is also used for intrinsic evaluation. Perplexity (PPL) is an easier-to-interpret transformation of cross-entropy in [1,|V|] where lower is better (less perplex):

$$\text{PPL}(X) = \text{exp}(-\frac{1}{N} \sum_{i} \text{log}(p(x_i \mid x_1, \ldots, x_{i-1})))$$

- The base depends on preference: e is efficient to compute, 2 (bits) stems from the definition of Shannon entropy, and 10 is easiest to calculate by hand.

QUESTIONS?

IMPROVING N-GRAM LANGUAGE MODELS

Smoothing: Redistribute probability mass from observed to rare or unobserved n-grams to improve generation.

Two situations for smoothing:

1. The count of the denominator is zero or very low.

$$p(\texttt{eggs} \mid \texttt{I like green}) = \frac{\mathsf{C}(\texttt{I like green eggs})}{\mathsf{C}(\texttt{I like green})}$$

2. The count of the numerator is zero or very low.

$$p(\texttt{ham} \mid \texttt{I do not like green}) = \frac{\mathsf{C}(\texttt{I do not like green ham})}{\mathsf{C}(\texttt{I do not like green})}$$

What is the problem in each situation?

SMOOTHING: “STUPID” BACKOFF

Use the n-gram probability of a model with a smaller τ instead.

$$p(x_n \mid x_{n-\tau}, \dots, x_{n-1}) \approx \cdots \approx p(x_n \mid x_{n-2}, x_{n-1}) \approx p(x_n \mid x_{n-1})$$

– Progressively reduce τ until the condition is sufficiently common (> λ)

$$p_{BO}(x_n \mid x_{n-\tau}, \dots, x_{n-1}) = \begin{cases} p(x_n \mid x_{n-\tau}, \dots, x_{n-1}) & \text{if } \mathrm{C}(x_{n-\tau}, \dots, x_{n-1}) > \lambda \\ p_{BO}(x_n \mid x_{n-(\tau-1)}, \dots, x_{n-1}) & \text{otherwise} \end{cases}$$

– λ can be set manually or inferred. $\lambda>1$ is roughly equivalent to $p(x_{n-\tau},\dots, x_{n-1})>0$.

SMOOTHING: ADD-ONE (LAPLACE)

Add 1 to the count of every n-gram before the likelihood estimation to avoid zero-probabilities.

$$p_{\text{add}-1}(x_n \mid x_{n-\tau}, \ldots, x_{n-1}) = \frac{1 + \mathrm{C}(x_{n-\tau}, \ldots, x_{n-1}, x_n)}{|V| + \sum_{x_i \in V} \mathrm{C}(x_{n-\tau}, \ldots, x_{n-1}, x_i)}$$

3-gram counts

3-gram	Sam	house	eggs	ham	box
i am	2	0	0	0	0
in a	0	8	0	0	7
like green	0	0	9	0	0

3-gram probabilities (MLE)

3-gram	Sam	house	eggs	ham	box
i am	1	0	0	0	0
in a	0	0.54	0	0	0.46
like green	0	0	1	0	0

3-gram counts (+1)

3-gram	Sam	house	eggs	ham	box
i am	3	1	1	1	1
in a	1	9	1	1	8
like green	1	1	10	1	1

3-gram probabilities (add-1)

3-gram	Sam	house	eggs	ham	box
i am	0.43	0.14	0.14	0.14	0.14
in a	0.05	0.40	0.05	0.05	0.35
like green	0.07	0.07	0.71	0.07	0.07

QUESTIONS?

ANNOUNCEMENT

Room change for next week (30.04.2026): Lecture in Hörsaal 8!